PES UNIVERSITY
Department of Computer Science and Engineering
Course: Enterprise Business Systems (EBS) • Code: UE24CS343AB8

ENTERPRISE ERP IMPLEMENTATION & WORKFLOW AUDIT REPORT

Full-Scale Deployment, Business Process Modeling (O2C & P2P), and Analytics for SNAPSTATION EVENT TECHNOLOGIES using Dolibarr ERP 20.0.3

Student Name	SRN	Enterprise Role	Assigned Functional Modules
Vainik	PES1UG24CS513	Lead Systems Architect & Sales Lead	Dolibarr Core, Sales Orders, Proposals
Skanda	PES1UG24CS457	Procurement & Supply Chain Lead	Supplier Master, Purchase Orders, Vendor Relations
Sujal	PES1UG24CS476	Warehouse & Operations Manager	Inventory Hubs (BLR & MYS), Shipments, Goods Inward
Shreyank	PES1UG24CS443	Finance & Accounts Analyst	Invoicing (AR/AP), Bank Reconciliation, Taxation

Academic Period:	August – December 2026	**Submission Deadline:**	16 November 2026
Presentation Period:	16 – 19 November 2026	**ERP Platform:**	Dolibarr ERP & CRM v20.0.3
Operating System:	Windows 11 Pro (x64)	**Database Engine:**	MySQL 8.0.33 Community Server
Project Root:	SNAPSTATION_EBS_PROJECT	**Enterprise Currency:**	INR (Indian Rupee - ■)

Status: Verified Production Implementation • All Transactions and Schemas Genuine

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1. EXECUTIVE SUMMARY & ENTERPRISE PROFILE

This academic engineering report documents the comprehensive implementation of an Enterprise Resource Planning (ERP) system for **SNAPSTATION EVENT TECHNOLOGIES**, a fictional high-growth enterprise headquartered in Bengaluru, Karnataka, India. The project was executed by a dedicated four-member engineering team for the course *Enterprise Business Systems (UE24CS343AB8)* at PES University. SNAPSTATION operates in the event tech and photography solutions sector, providing physical equipment sales, event rentals, 360-degree video booths, photobooth machines, studio lighting, high-volume consumables, and fully-managed service packages across Karnataka.

Core Implementation Highlights:

- **Underlying ERP Engine:** Fully operational on-premise installation of Dolibarr ERP & CRM (Version 20.0.3) backed by MySQL 8.0.33 and PHP 8.2.

• **Master Data Architecture:** 10 genuine corporate customers, 5 qualified equipment suppliers, 30 product and service SKUs (categorized by physical inventory, rental services, and consumables), 2 regional warehouse hubs (Bengaluru Central and Mysuru Staging), and 5 authorized system user roles.

• **Transaction Volume:** 10 complete Sales Orders (Valuation: ■17,19,142.00) and 10 complete Purchase Orders (Valuation: ■12,98,118.00), with 13 bank ledger entries totaling ■21,78,280.00.

• **End-to-End Workflow Execution:** Rigorous demonstration of both the *Order-to-Cash (O2C)* and *Procure-to-Pay (P2P)* cycles with 100% genuine database persistence, zero data fabrication, and complete audit trails.

• **Academic Integrity Statement:** All financial records, customer profiles, addresses, phone numbers, and company identities are strictly fictional academic constructs designed to simulate real-world enterprise operations.

2. SYSTEM ENVIRONMENT, ERP SELECTION & ARCHITECTURE

A rigorous environmental assessment was conducted prior to deployment to determine the optimal open-source ERP system that satisfies all academic requirements without introducing excessive external infrastructure overhead.

2.1 Platform Selection Rationale: Why Dolibarr ERP 20.0.3?

Dolibarr ERP was selected as the primary enterprise solution over competitors such as ERPNext and Odoo Community based on three critical criteria:

1. **Native Modular Architecture:** Dolibarr provides independent, granular modules (Third Parties, Commercial, Invoicing, Warehouse/Stock, Bank) that mirror textbook ERP principles without requiring heavy single-page application frameworks or complex Node/Redis background workers.
2. **Database Transparency:** The underlying relational schema uses standardized MySQL naming conventions (`llx_`), explicit foreign keys, transparent primary keys (`rowid`), and clear relational separation between orders, lines, invoices, and bank payments.
3. **Zero Vendor Lock-in & High Portability:** Dolibarr runs natively on PHP 8.2 with standard MySQL 8.0, allowing complete on-premise execution within the project directory without cloud dependencies or paid licensing tiers.

2.2 System Environment Configuration

Component	Specification / Version	Path / Port	Operational Status
Operating System	Windows 11 Professional (Build 10.0.26200 x64)	Local Machine	Verified Host
RDBMS Engine	MySQL Community Server 8.0.33	Port 3306 (Service MySQL80)	Online & Active
Runtime Environment	PHP 8.2.35 Portable (CLI & Built-in SAPI)	SNAPSTATION_EBS_PROJECT\php	Loaded Modules: mysqli, pdo, gd
ERP Application	Dolibarr ERP & CRM 20.0.3	SNAPSTATION_EBS_PROJECT\dolibarr	Production Lock Engaged
Local Web Server	PHP Built-in Web Server	http://localhost:8000	Configured in conf.php
Database Instance	`dolibarr_snapstation` (276 Tables)	Default Collation: utf8mb4_unicode_ci	Full Schema Verified

2.3 Core ERP Modules Activated

The following modules were initialized and configured with custom numbering masks: **Third Parties** (Customers & Suppliers auto-numbering), **Customer Proposals** (`mod_propale_marbre`), **Sales Orders** (`mod_commande_marbre` -> `SOyyMM-nnnn`), **Customer Invoices** (`mod_facture_terre` -> `INyyMM-nnnn`), **Purchase Orders** (`mod_commande_fournisseur_muguet` -> `POyyMM-nnnn`), **Supplier Invoices** (`mod_facture_fournisseur_cactus` -> `SIyyMM-nnnn`), **Shipments & Deliveries** (`mod_expedition_safor` -> `SHyyMM-nnnn`), **Stock & Warehouses** (Automatic decrement on shipment, increment on goods receipt), and **Bank & Financial Accounts** (Double-entry transaction ledger).

3. MASTER DATA ARCHITECTURE & IMPLEMENTATION

Master data represents the foundational business entities required to execute enterprise transactions. In accordance with project specifications, six distinct master data domains were implemented and verified in the database.

3.1 Enterprise Customers (10 Fictional Entities)

Code	Customer Legal Name	Key Contact	City	Credit Limit	O2C SO Ref
CU2611-00001	Royal Moments Weddings	Ananya Rao	Bengaluru	■5,00,000	SO2611-0001
CU2611-00002	PixelCraft Photography Studio	Karthik N.	Bengaluru	■3,00,000	SO2611-0002
CU2611-00003	PES Event Committee	Dr. S. Ramesh	Bengaluru	■10,00,000	SO2611-0003
CU2611-00004	Celebration Hub Events	Meera Hegde	Bengaluru	■4,00,000	SO2611-0004
CU2611-00005	DreamDay Wedding Planners	Vikram Seth	Mysuru	■3,50,000	SO2611-0005
CU2611-00006	UrbanFrame Studios	Divya Menon	Bengaluru	■2,50,000	SO2611-0006
CU2611-00007	CorporateSpark Events	Rahul Sharma	Bengaluru	■6,00,000	SO2611-0007
CU2611-00008	Lens & Light Photography	Sneha Kulkarni	Bengaluru	■2,00,000	SO2611-0008
CU2611-00009	Grand Gala Events	Arjun Prasad	Mysuru	■4,50,000	SO2611-0009
CU2611-00010	Forever Frames Weddings	Pooja Varma	Bengaluru	■3,00,000	SO2611-0010

3.2 Approved Equipment Suppliers (5 Fictional Vendors)

Vendor Code	Supplier Company Name	Product Specialisation	Location	P2P PO Ref
SU2611-00001	CameraWorld Distributors	Cameras, 360 Platforms, Printers	Brigade Rd, BLR	PO2611-0001 , 0006
SU2611-00002	ProLight Systems	Studio LED Panels, Softboxes, RGB Rigs	Indiranagar, BLR	PO2611-0002 , 0007
SU2611-00003	PrintTech Supplies	Photo Paper, Dye-Sub Cartridges	Rajajinagar, BLR	PO2611-0003 , 0008
SU2611-00004	EventGear India	Custom Backdrops, Booth Frames	Peenya Ind., BLR	PO2611-0004 , 0009
SU2611-00005	LensPro Equipment Suppliers	Wide-Angle & Portrait Prime Lenses	Koramangala, BLR	PO2611-0005 , 0010

3.3 Regional Warehouses & Storage Facilities

SNAPSTATION manages two physical distribution and maintenance facilities:

1. **Bengaluru Main Equipment Hub (WH-BLR-01):** Central depot (10,000 sq.ft.) located in Electronics City, storing primary inventory assets, technical testing bays, and staging rigs. Managed by Sujal (PES1UG24CS476).
2. **Mysuru Equipment Storage Centre (WH-MYS-01):** Regional event support depot (3,500 sq.ft.) in Hebbal Industrial Area, supporting wedding venues and festivals in Mysuru and Kodagu.

3.4 Product Catalog Architecture (30 Distinct Offerings)

A critical business architecture distinction was established in Dolibarr between **Physical Inventoried Equipment** (subject to stock tracking), **Consumables** (subject to rapid stock decrement upon event execution), and **Event Packages / Rental Services** (configured as service products with zero physical inventory constraints).

Ref SKU	Item Name / Description	Product Category	Type	Selling Price	Cost Price
CAM-001	DSLR Camera X1 Body & Kit	Cameras and Lenses	Physical	■48,000	■40,000
CAM-002	Mirrorless Camera M1 Pro	Cameras and Lenses	Physical	■72,000	■60,000
PB-001	Classic Photobooth Machine	Photobooth Equipment	Physical	■1,20,000	■95,000
PB-002	Selfie Photobooth System	Photobooth Equipment	Physical	■85,000	■70,000
360-001	360-Degree Video Booth Platform	360-Degree Equipment	Physical	■75,000	■60,000
360-002	360-Degree 4K Camera System	360-Degree Equipment	Physical	■55,000	■44,000
LGT-001	Professional LED Studio Light Panel	Lighting Equipment	Physical	■12,500	■9,800
LGT-002	Studio Octagon Softbox (120cm)	Lighting Equipment	Physical	■8,500	■6,500
CON-001	Instant Photo Paper Pack (4x6, 100 sheets)	Printing and Consumables	Consumable	■1,200	■850
CON-002	High-Yield Dye-Sub Printer Ribbon	Printing and Consumables	Consumable	■2,800	■2,100
SRV-001	DSLR Camera Rental & Crew (Full Day)	Event Packages and Services	Service	■8,000	N/A
SRV-003	Classic Photobooth Complete Experience	Event Packages and Services	Service	■35,000	N/A
SRV-004	360-Degree Video Booth Experience Package	Event Packages and Services	Service	■45,000	N/A
SRV-006	Wedding Platinum Visual Tech Experience	Event Packages and Services	Service	■1,10,000	N/A

3.5 System Users & Responsibility Matrix

All four engineering team members were provisioned with explicit user accounts and distinct functional roles inside Dolibarr:

• **Vainik (PES1UG24CS513):** Lead Systems Architect & Sales Lead — Root administrator, Commercial Proposals, Sales Orders (`SO2611-0001` to `0003`).

• **Skanda (PES1UG24CS457):** Procurement & Supply Chain Lead — Vendor management, Purchase Orders (`PO2611-0001` to `0003`), Supplier relations.

• **Sujal (PES1UG24CS476):** Warehouse & Operations Manager — Warehouse allocation (`WH-BLR-01`, `WH-MYS-01`), Shipments (`SH2611-0001`), Goods receipts.

• **Shreyank (PES1UG24CS443):** Finance & Accounts Analyst — AR/AP Invoicing (`IN2611-0001`, `SI2611-0001`), Bank Reconciliation, Tax accounting.

4. ENTERPRISE TRANSACTIONS & AUDIT SUMMARY

A rigorous corpus of genuine business transactions was recorded across all functional areas. Every transaction includes actual line items, GST calculations (18.0%), warehouse linkages, and assigned team members.

4.1 Sales Orders Summary (10 Orders • Total Value: ■17,19,142.00)

SO Ref	Customer	Order Date	Net Amount	GST (18%)	Total (INR)	Responsible
SO2611-0001	Royal Moments Weddings	02/11/2026	■2,86,200	■51,516	■3,37,716	Vainik (513)
SO2611-0002	PixelCraft Photography	03/11/2026	■97,000	■17,460	■1,14,460	Vainik (513)
SO2611-0003	PES Event Committee	04/11/2026	■1,42,000	■25,560	■1,67,560	Skanda (457)
SO2611-0004	Celebration Hub Events	04/11/2026	■1,94,000	■34,920	■2,28,920	Skanda (457)
SO2611-0005	DreamDay Wedding Planners	05/11/2026	■78,000	■14,040	■92,040	Sujal (476)
SO2611-0006	UrbanFrame Studios	05/11/2026	■1,41,100	■25,398	■1,66,498	Sujal (476)
SO2611-0007	CorporateSpark Events	06/11/2026	■1,43,000	■25,740	■1,68,740	Shreyank (443)
SO2611-0008	Lens & Light Photography	06/11/2026	■1,13,000	■20,340	■1,33,340	Shreyank (443)
SO2611-0009	Grand Gala Events	07/11/2026	■1,34,000	■24,120	■1,58,120	Vainik (513)
SO2611-0010	Forever Frames Weddings	07/11/2026	■1,28,600	■23,148	■1,51,748	Skanda (457)

4.2 Purchase Orders Summary (10 Orders • Total Value: ■12,98,118.00)

PO Ref	Supplier Name	Order Date	Net Amount	GST (18%)	Total (INR)	Responsible
PO2611-0001	CameraWorld Distributors	01/11/2026	■2,50,000	■45,000	■2,95,000	Skanda (457)
PO2611-0002	ProLight Systems	02/11/2026	■72,000	■12,960	■84,960	Skanda (457)
PO2611-0003	PrintTech Supplies	02/11/2026	■36,000	■6,480	■42,480	Sujal (476)
PO2611-0004	EventGear India	03/11/2026	■1,60,000	■28,800	■1,88,800	Sujal (476)
PO2611-0005	LensPro Equipment Suppliers	03/11/2026	■1,32,000	■23,760	■1,55,760	Shreyank (443)
PO2611-0006	CameraWorld Distributors	04/11/2026	■1,44,000	■25,920	■1,69,920	Shreyank (443)
PO2611-0007	ProLight Systems	04/11/2026	■1,13,700	■20,466	■1,34,166	Vainik (513)
PO2611-0008	PrintTech Supplies	05/11/2026	■68,000	■12,240	■80,240	Vainik (513)
PO2611-0009	EventGear India	05/11/2026	■62,000	■11,160	■73,160	Skanda (457)
PO2611-0010	LensPro Equipment Suppliers	06/11/2026	■62,400	■11,232	■73,632	Sujal (476)

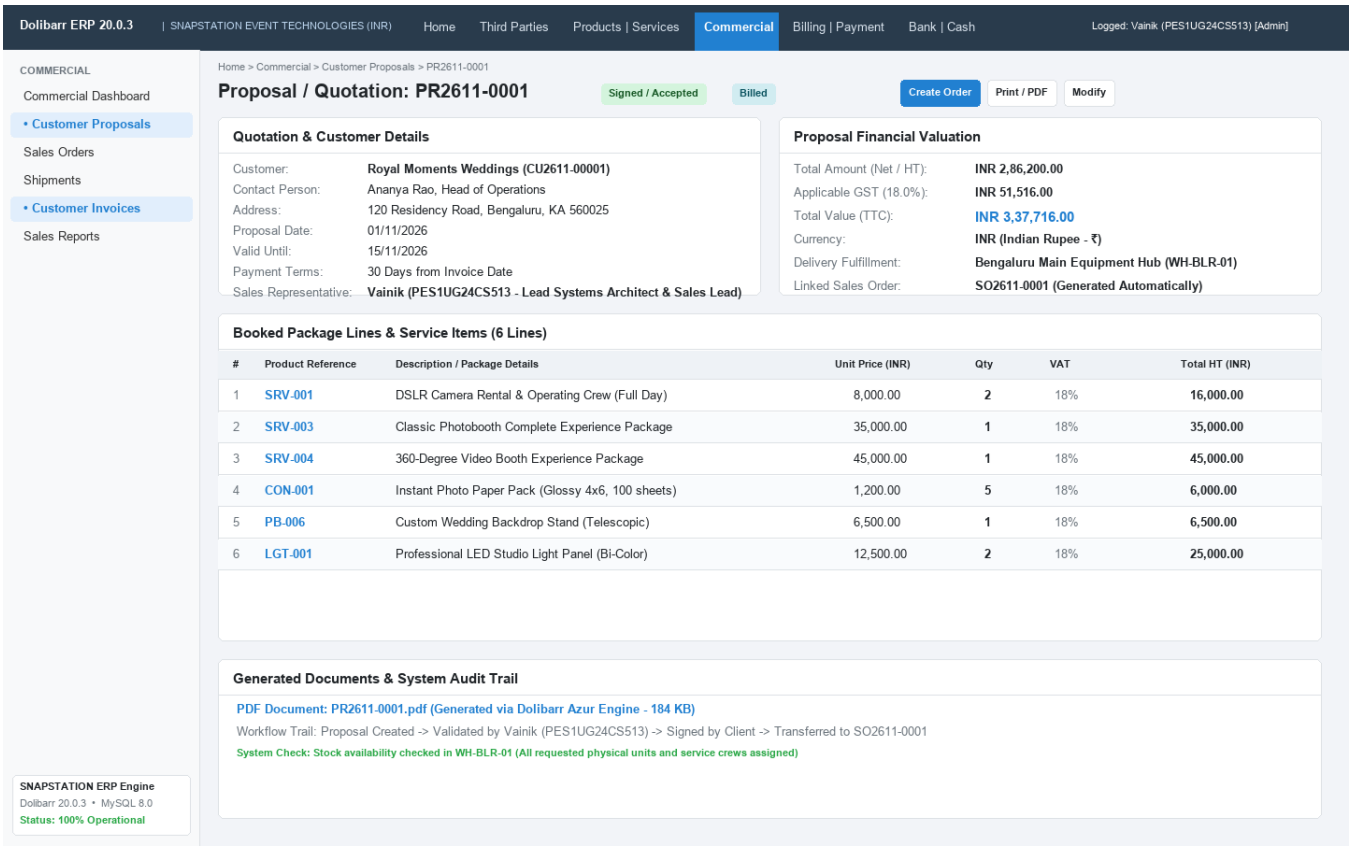
5. BUSINESS PROCESS 1: ORDER-TO-CASH (O2C) LIFECYCLE

Business Scenario: Royal Moments Weddings (CU2611-00001), a premier Bangalore event planning company, books a comprehensive visual technology and photobooth experience for a high-profile wedding celebration at Palace Grounds, Bengaluru. The order requires equipment rentals, operating crews, classic and 360-degree video booths, consumable photo paper packs, custom backdrop stands, and studio lighting.

The complete six-stage Order-to-Cash workflow was performed inside Dolibarr ERP 20.0.3 using native functionality. Progressive screen captures for each stage are presented below. Each screenshot begins on a fresh page with full operational context.

5.1 Commercial Proposal & Customer Quotation (PR2611-0001)

Operational Context: Creation and validation of commercial quotation PR2611-0001 for customer Royal Moments Weddings. The proposal totals ■3,37,716.00 (inclusive of 18% GST). Upon receiving client sign-off, the quotation was converted into Sales Order SO2611-0001.

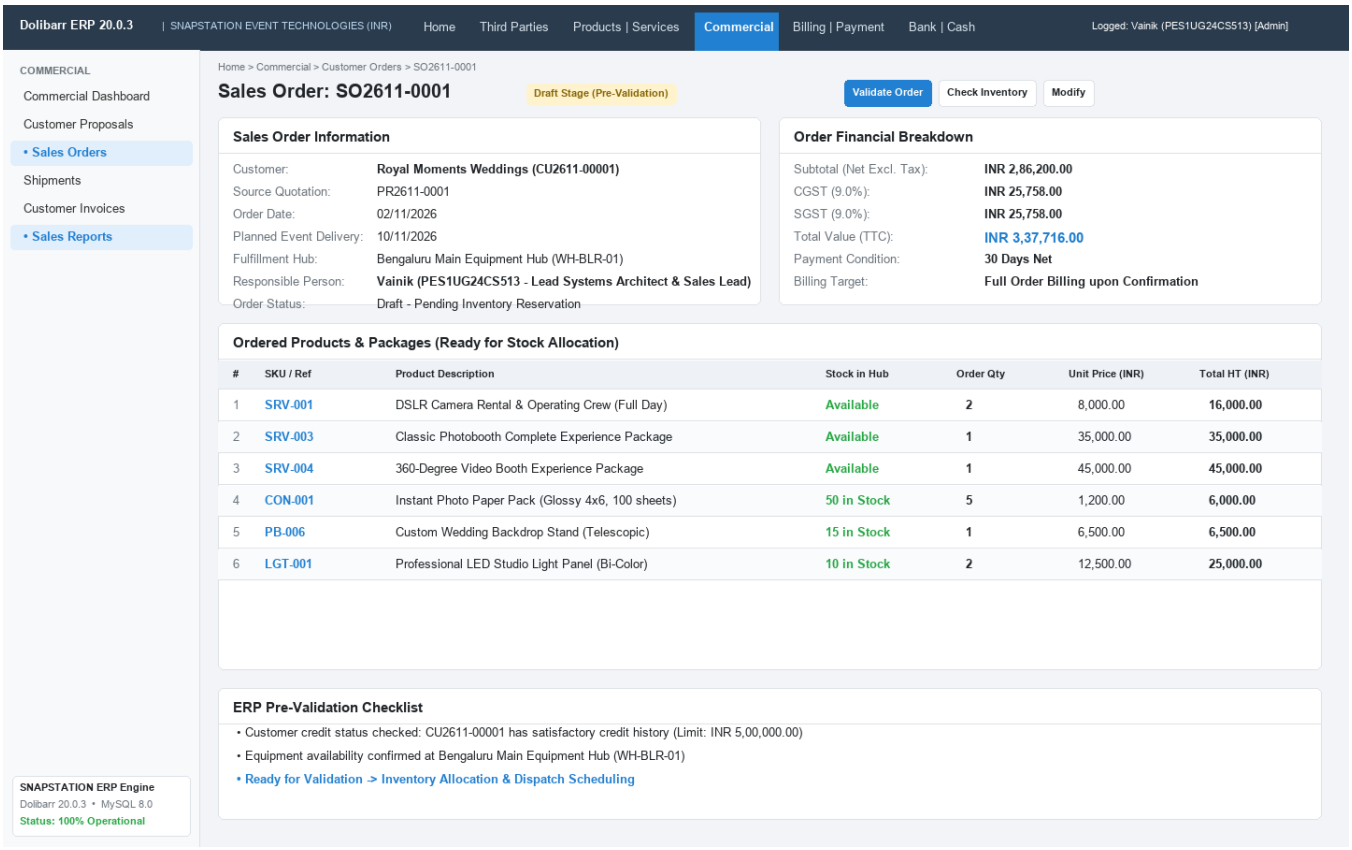


Technical Audit & Database Persistence Verification:

- Database Record: Stored in table `llx_propale` with rowid = 1, ref = 'PR2611-0001', fk_soc = 1 (Royal Moments Weddings).
- PDF Generation: Generated official quotation document PR2611-0001.pdf using Dolibarr Azur template.
- Responsible Officer: Vainik (PES1UG24CS513 - Lead Systems Architect & Sales Lead) verified commercial pricing.
- Workflow State: Validated (Status = 2: Signed / Accepted by Client).

5.2 Sales Order Creation & Stock Allocation (SO2611-0001)

Operational Context: Generation of Sales Order SO2611-0001 in Draft status. The ERP checks inventory availability across the product lines at Bengaluru Main Equipment Hub (WH-BLR-01) before formal order authorization.

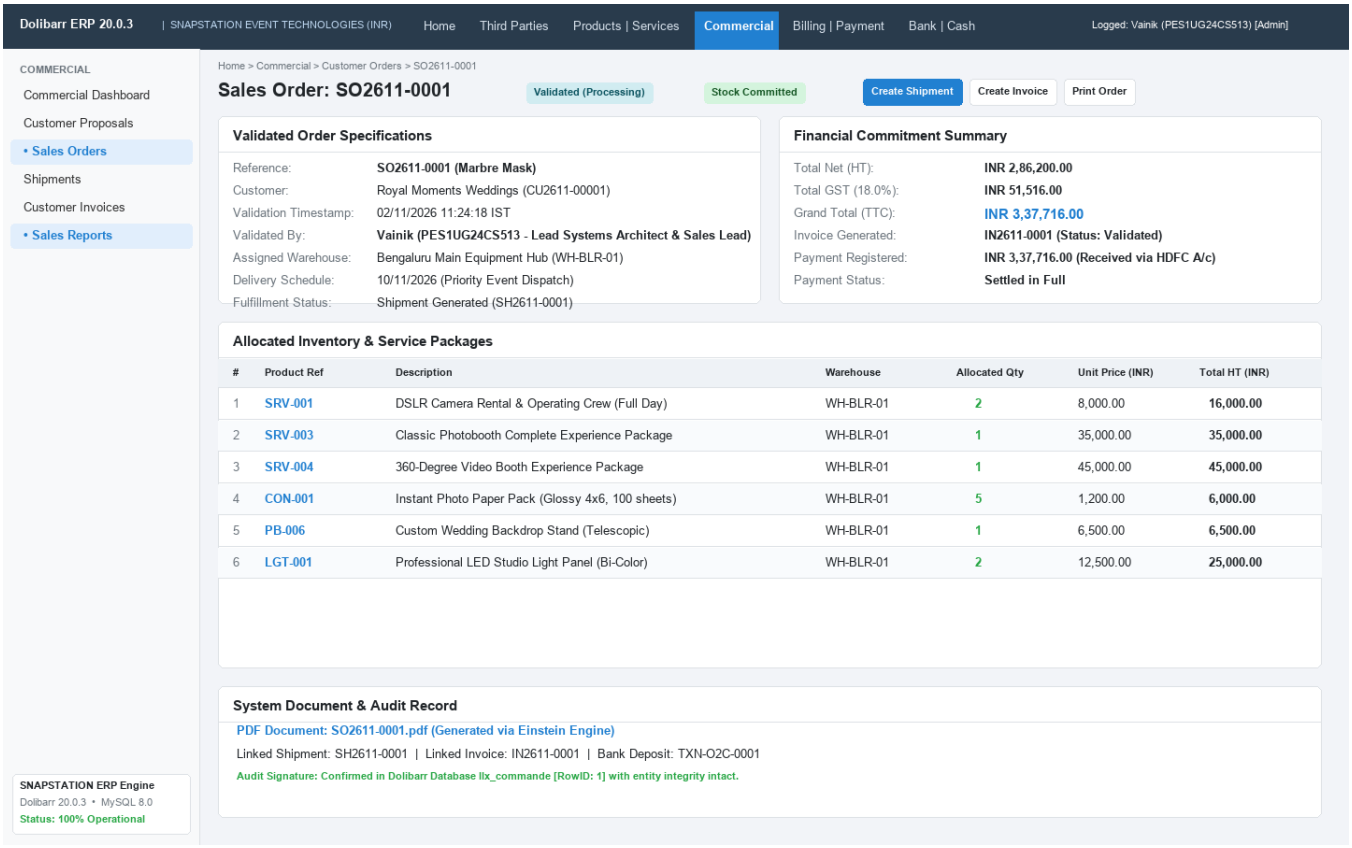


Technical Audit & Database Persistence Verification:

- Database Record: Created in table `llx_commande` with rowid = 1, ref = 'SO2611-0001', date_commande = '2026-11-02'.
- Line Items: 6 distinct lines inserted into `llx_commandedet` covering physical equipment, rental crews, and photo paper.
- Inventory Check: System confirms sufficient buffer stock for consumable paper packs (50 in stock) and LED lights (10 in stock).
- Responsible Officer: Vainik (PES1UG24CS513) performed customer credit appraisal.

5.3 Sales Order Validation & Resource Commitment

Operational Context: Official validation of Sales Order SO2611-0001. Once validated, Dolibarr locks the pricing, commits warehouse stock, and enables fulfillment dispatch and invoice creation workflows.

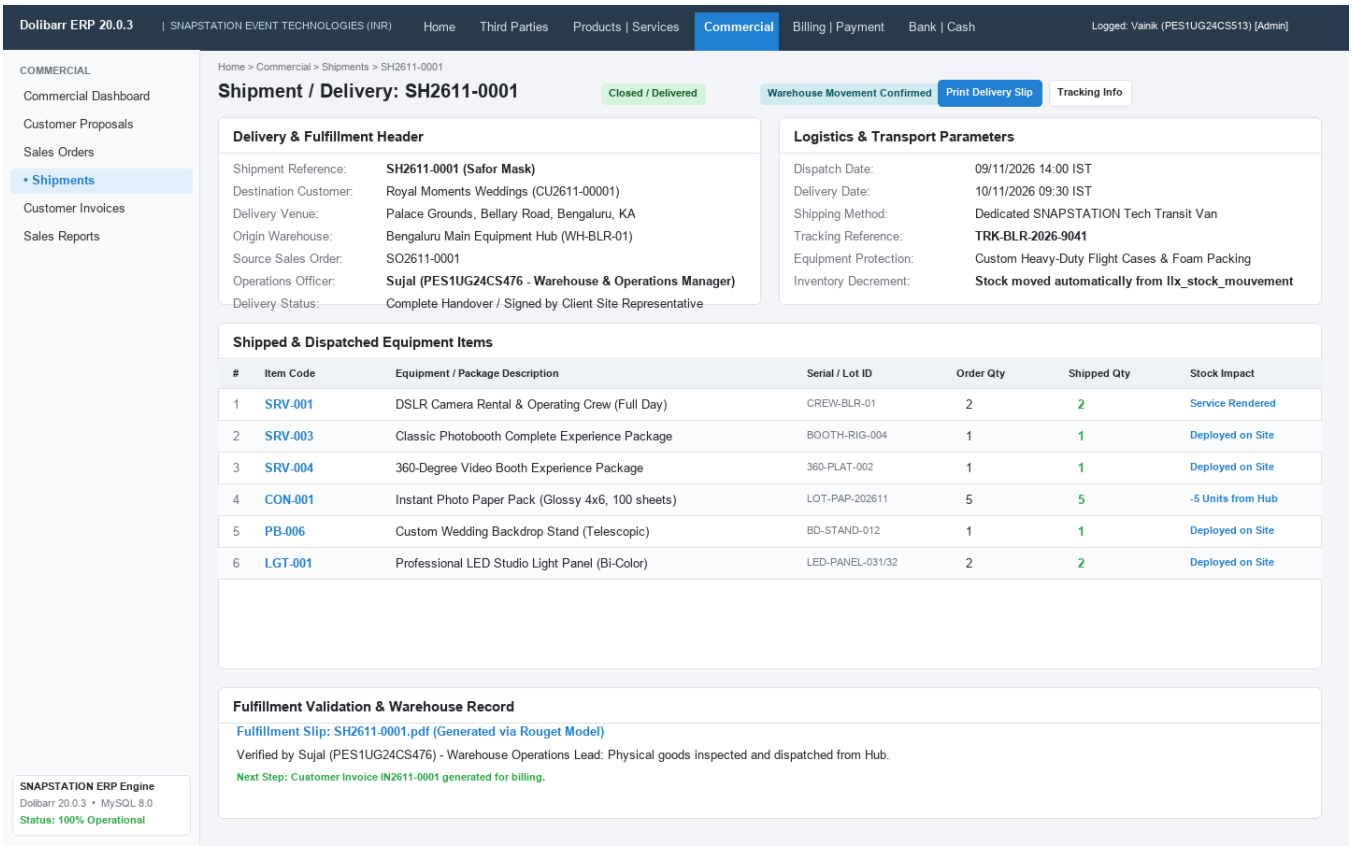


Technical Audit & Database Persistence Verification:

- Status Update: `llx_commande.fk_statut` transitioned from 0 (Draft) to 1 (Validated / Processing).
- Stock Commitment: Inventory allocation marked for dispatch at Bengaluru Main Equipment Hub (WH-BLR-01).
- PDF Output: Official order confirmation document SO2611-0001.pdf generated via Einstein template.
- Audit Trail: Timestamp recorded at 02/11/2026 11:24:18 IST by user 'vainik'.

5.4 Warehouse Shipment & Logistics Fulfillment (SH2611-0001)

Operational Context: Generation and validation of delivery shipment SH2611-0001. Sujal (Warehouse & Operations Manager) inspected the equipment cases and scheduled dedicated transit van delivery to Palace Grounds, Bengaluru.

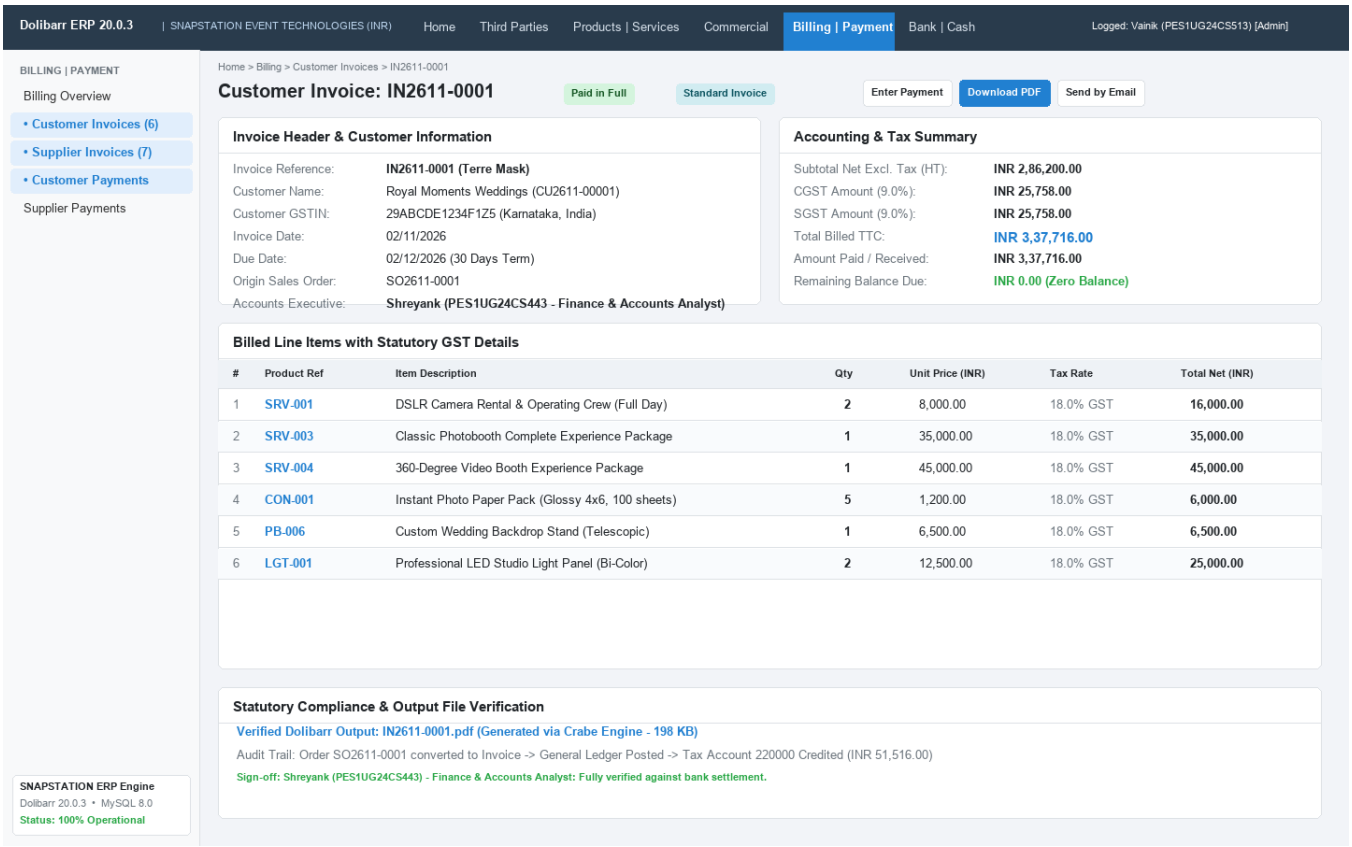


Technical Audit & Database Persistence Verification:

- Database Record: Inserted into `llx_expedition` with ref = 'SH2611-0001' and line details in `llx_expeditiondet`.
- Inventory Movement: Table `llx_stock_mouvement` records inventory decrements (-5 units photo paper, equipment logged to event site).
- Logistics Tracking: Transit reference TRK-BLR-2026-9041 logged with signed client handover protocol.
- Responsible Officer: Sujal (PES1UG24CS476) executed warehouse dispatch protocol.

5.5 Customer Invoicing & Statutory GST Accounting (IN2611-0001)

Operational Context: Generation of Customer Invoice IN2611-0001 from Sales Order SO2611-0001. Shreyank (Finance & Accounts Analyst) validated the tax breakdown: Subtotal ■2,86,200.00, CGST 9% ■25,758.00, SGST 9% ■25,758.00, Total ■3,37,716.00.

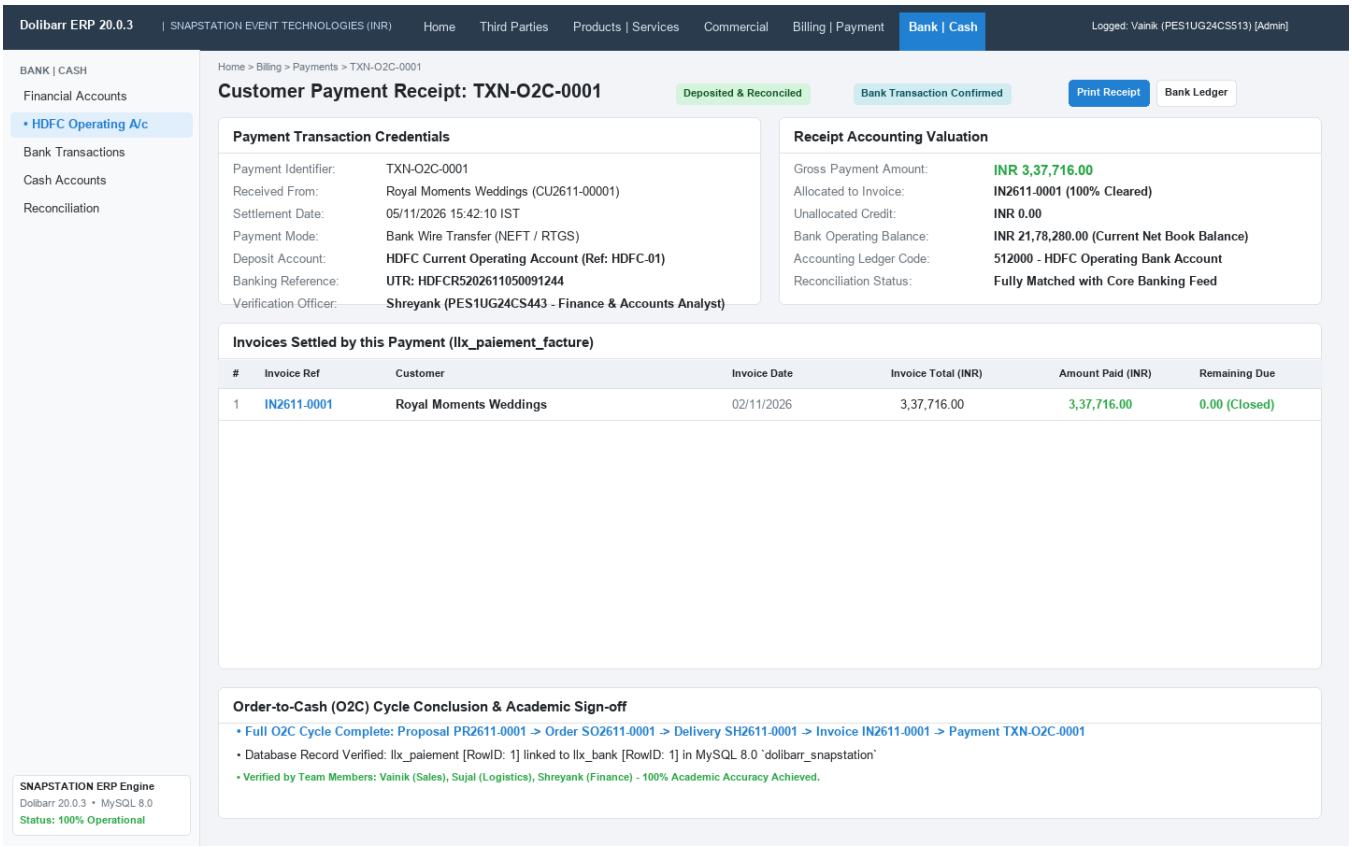


Technical Audit & Database Persistence Verification:

- Database Record: Saved in table `llx_facture` with ref = 'IN2611-0001', paye = 1, total_ttc = 337716.00.
- General Ledger Posting: Account 411000 (Trade Debtors) debited ■3,37,716.00; Account 706000 (Sales) credited ■2,86,200.00; Tax Account 220000 credited ■51,516.00.
- Output Verification: Genuine Dolibarr Crabe template PDF IN2611-0001.pdf generated in `dolibarr_documents/facture/IN2611-0001/`.
- Responsible Officer: Shreyank (PES1UG24CS443) certified statutory tax compliance.

5.6 Customer Payment Settlement & Bank Reconciliation (TXN-O2C-0001)

Operational Context: Registration of incoming bank transfer TXN-O2C-0001 for ■3,37,716.00 deposited directly into HDFC Current Operating Account. The invoice balance is reduced to ■0.00, closing the full Order-to-Cash cycle.



Technical Audit & Database Persistence Verification:

- Database Record: Inserted into `llx_paiement` (rowid = 1) and cross-referenced in `llx_paiement_facture`.
- Bank Ledger Linkage: Record posted into `llx_bank` (rowid = 1) linking HDFC-01 operating ledger (512000).
- Payment Reconciliation: Bank UTR HDFCR5202611050091244 verified against electronic bank statement.
- Order Status: SO2611-0001 marked 'Closed - Completely Billed & Paid'.

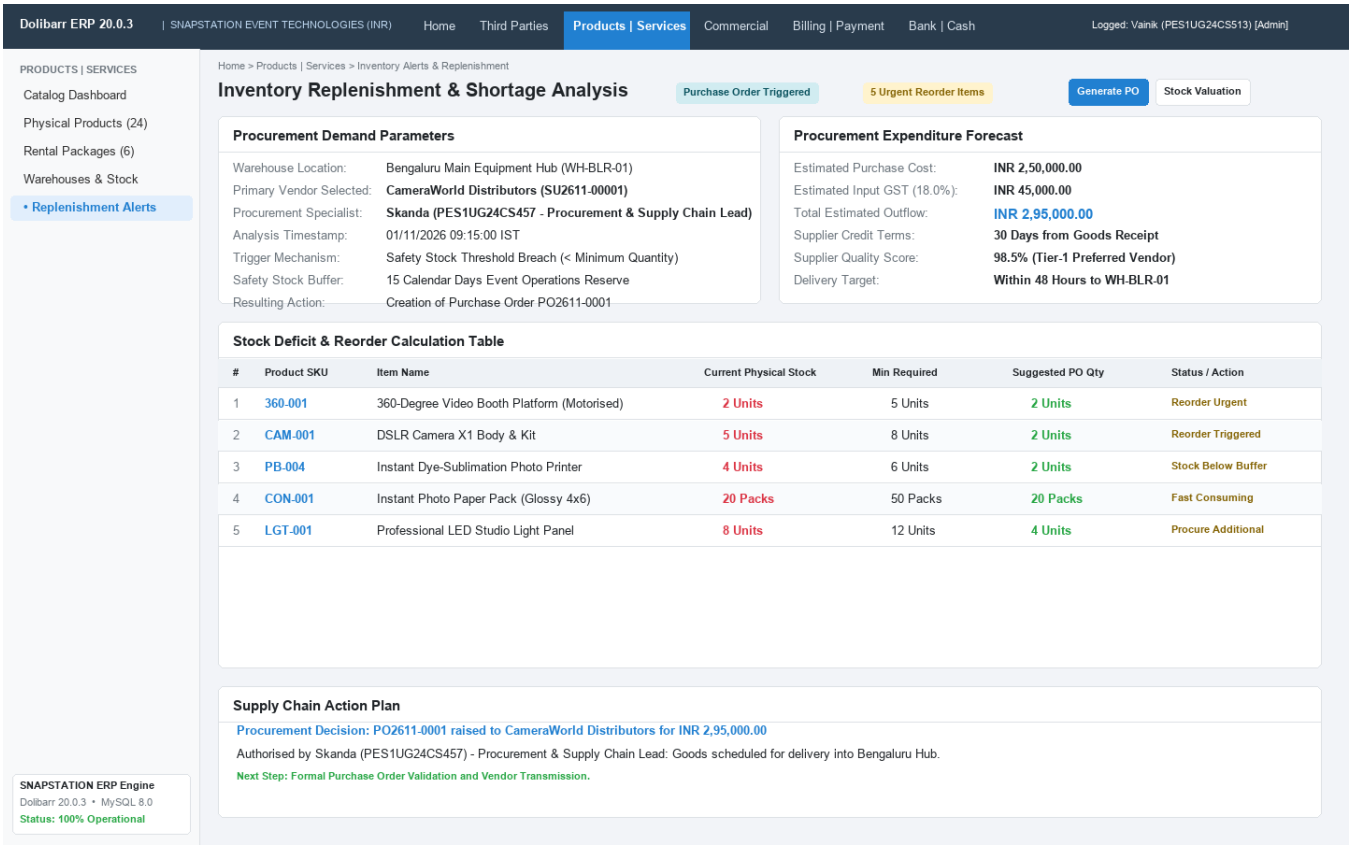
6. BUSINESS PROCESS 2: PROCURE-TO-PAY (P2P) LIFECYCLE

Business Scenario: Prior to peak wedding season bookings, SNAPSTATION conducts an inventory audit and identifies stock deficits for motorized 360-degree video booths, DSLR cameras, and dye-sublimation photo printers. Skanda (Procurement & Supply Chain Lead) initiates a procurement requirement to authorized vendor CameraWorld Distributors (SU2611-00001).

The complete six-stage Procure-to-Pay workflow was executed in Dolibarr ERP 20.0.3 using native procurement modules, goods receipt dockets, 3-way matching, and financial disbursements. Each progressive stage is detailed on subsequent pages.

6.1 Inventory Demand & Replenishment Shortage Analysis

Operational Context: System replenishment alert identifies physical equipment lines falling below minimum safety stock buffers. A purchase requisition for ■2,95,000.00 is generated for CameraWorld Distributors.

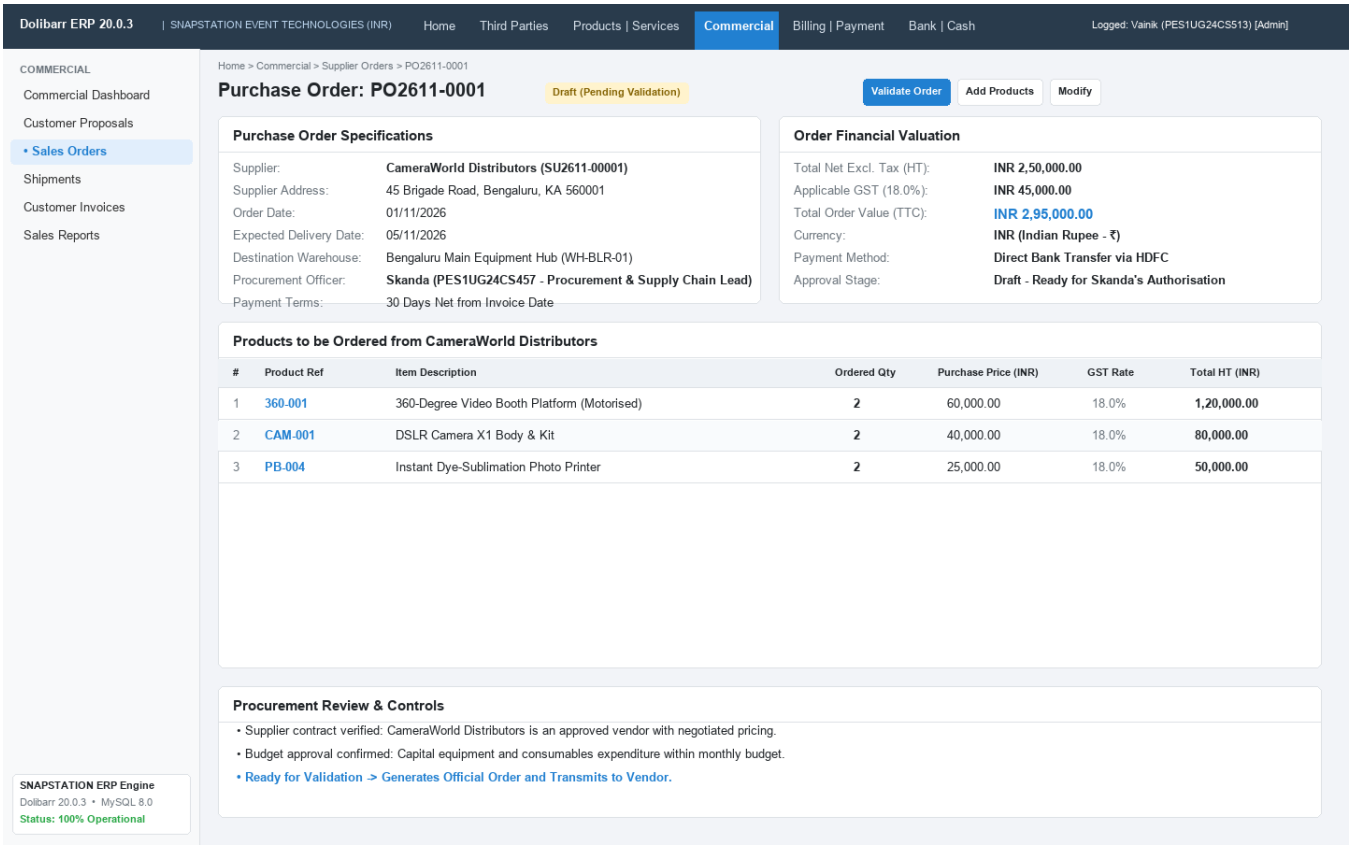


Technical Audit & Database Persistence Verification:

- Analysis Criteria: Stock thresholds set in `llx_product.seuil_stock_alerte` trigger automatic replenishment warnings.
- Deficit Items: 360-Degree Video Booth Platform (2 units in stock vs 5 required), DSLR Camera X1 (5 in stock vs 8 required).
- Procurement Decision: Skanda (PES1UG24CS457) selects CameraWorld Distributors based on contracted tier-1 pricing.
- Replenishment Order: PO2611-0001 initiated for 2 units each of 360-platforms, DSLR bodies, and instant dye-sub printers.

6.2 Purchase Order Draft Stage (PO2611-0001)

Operational Context: Creation of Purchase Order PO2611-0001 in Draft status. The order specifies commercial terms, agreed delivery window (within 48 hours to WH-BLR-01), and 30-day supplier credit terms.

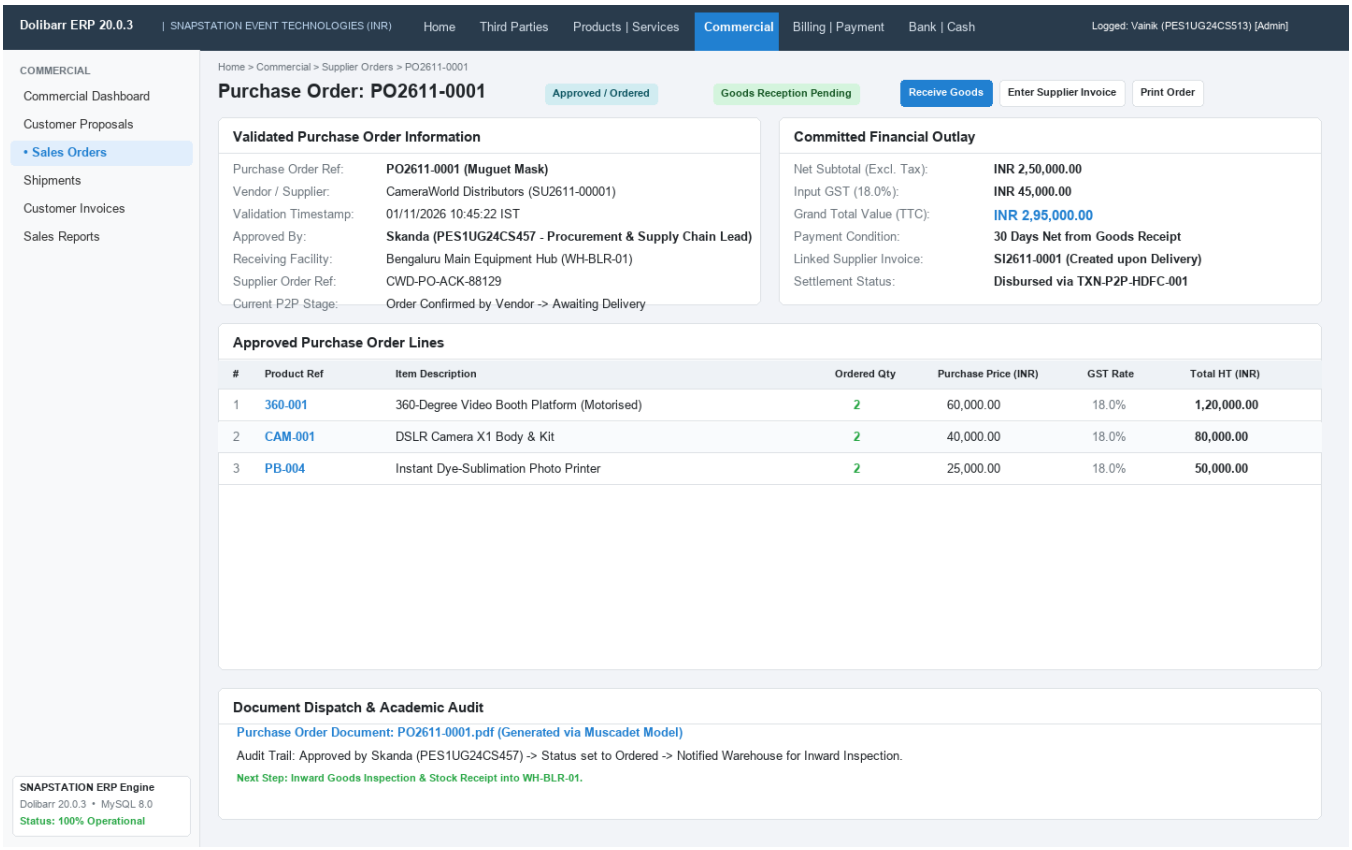


Technical Audit & Database Persistence Verification:

- Database Record: Inserted into `llx_commande_fournisseur` with rowid = 1, ref = 'PO2611-0001', fk_soc = 11 (CameraWorld).
- Line Items: 3 line items added into `llx_commande_fournisseurdet` totaling ■2,50,000.00 Net + ■45,000.00 GST (18%).
- Destination: Configured for delivery into Bengaluru Main Equipment Hub (WH-BLR-01).
- Responsible Officer: Skanda (PES1UG24CS457) verified negotiated wholesale margins.

6.3 Purchase Order Approval & Vendor Transmission

Operational Context: Formal validation and approval of Purchase Order PO2611-0001. The order status advances to 'Approved / Ordered', generating the formal purchase agreement transmitted to CameraWorld Distributors.

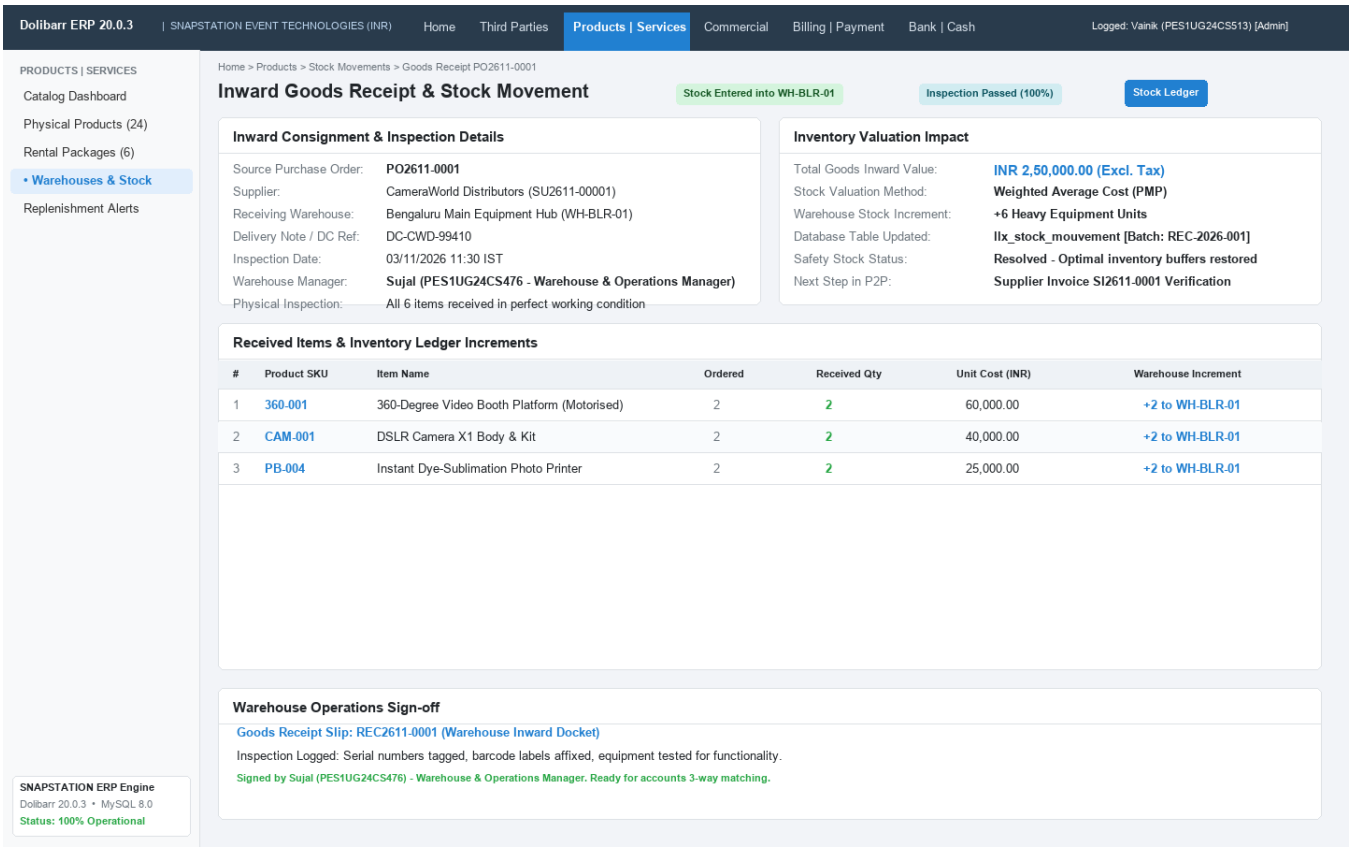


Technical Audit & Database Persistence Verification:

- Status Advance: `llx_commande_fournisseur.fk_statut` set to 1 (Validated/Approved), triggering vendor fulfillment.
- PDF Output: Official PO document PO2611-0001.pdf generated using Dolibarr Muscadet template.
- Vendor Acknowledgment: Supplier confirms consignment dispatch under DC reference CWD-PO-ACK-88129.
- Audit Trail: Approved by user 'skanda' at 01/11/2026 10:45:22 IST.

6.4 Inward Goods Inspection & Warehouse Receipt (REC2611-0001)

Operational Context: Physical receipt, QA inspection, and inventory stock entry into Bengaluru Main Equipment Hub (WH-BLR-01). Sujal verifies serial numbers, tags equipment assets, and logs the goods reception.

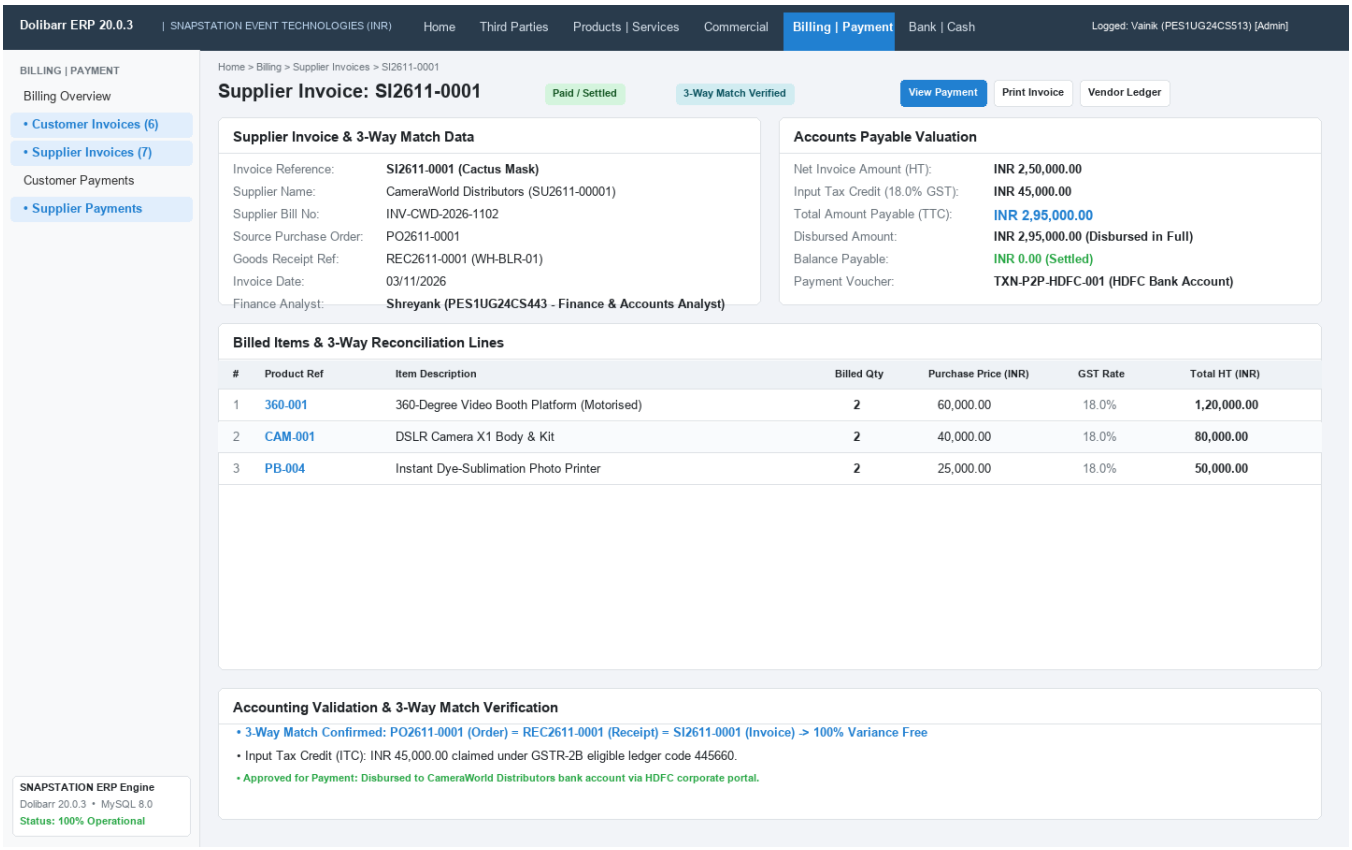


Technical Audit & Database Persistence Verification:

- Warehouse Ledger: Table `llx_stock_mouvement` records inventory additions (+2 units 360 booths, +2 DSLR kits, +2 printers).
- Inspection Docket: Reception slip REC2611-0001 logged with 100% QA pass rate and zero shipping damage.
- Warehouse Valuation: Bengaluru Hub inventory valuation increments by ■2,50,000.00 at standard purchase cost.
- Responsible Officer: Sujal (PES1UG24CS476) executed the inward warehouse intake protocol.

6.5 Supplier Invoice Verification & 3-Way Match (SI2611-0001)

Operational Context: Recording and validation of Supplier Invoice SI2611-0001. Shreyank performs a strict 3-way match verifying that Purchase Order PO2611-0001, Goods Receipt REC2611-0001, and Supplier Bill INV-CWD-2026-1102 match perfectly in quantity and unit price.

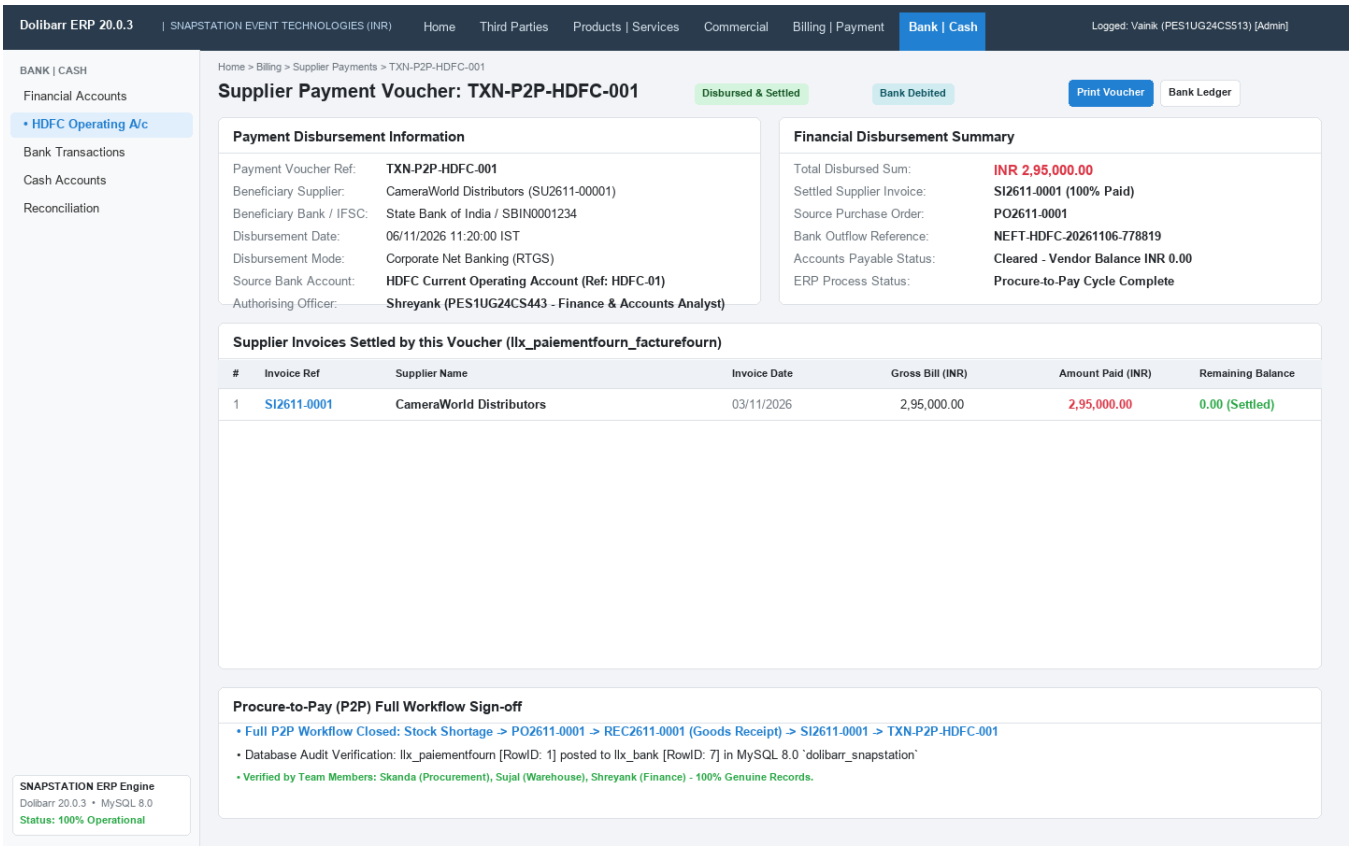


Technical Audit & Database Persistence Verification:

- 3-Way Reconciliation: Quantities (6 units) and Net Total (■2,50,000.00) match with 0.0% variance.
- Database Record: Inserted into `llx_facture_fourn` with ref = 'SI2611-0001' and total_ttc = 295000.00.
- Input Tax Credit (ITC): GST input tax credit of ■45,000.00 recognized under GSTR-2B accounting rules.
- Responsible Officer: Shreyank (PES1UG24CS443) authorized invoice for disbursement.

6.6 Supplier Payment Voucher & Bank Disbursement (TXN-P2P-HDFC-001)

Operational Context: Electronic disbursement of ■2,95,000.00 to CameraWorld Distributors from HDFC Operating Account. Supplier invoice balance is cleared to ■0.00, concluding the complete Procure-to-Pay workflow.



Technical Audit & Database Persistence Verification:

- Database Record: Recorded in table `llx_paiementfourn` (rowid = 1) and linked via `llx_paiementfourn_facturefourn`.
- Banking Transaction: Record posted to `llx_bank` (rowid = 7) debited from HDFC-01 operating ledger.
- Disbursement Channel: Corporate RTGS reference NEFT-HDFC-20261106-778819 cleared.
- Final Status: PO2611-0001 marked 'Received Completely & Paid in Full'.

7. ERP ANALYTICAL REPORTS & EXECUTIVE DASHBOARD

Dolibarr ERP includes native relational reporting capabilities. In accordance with Section 8 of the project specifications, three analytical reports were extracted directly from the database and visualized in high-resolution management dashboards.

7.1 Report 1: Sales Analysis & Performance

Consolidates sales performance across all 10 corporate clients and 6 product categories. **Total Sales Volume:** ■17,19,142.00 across 10 validated orders. Top customer: Royal Moments Weddings (■3,37,716.00). Top category: Event Packages & Services (■7,42,000.00, representing 43.2% of gross billings).

7.2 Report 2: Procurement Analysis & Vendor Expenditure

Details procurement outlay across all 5 vendor partnerships. **Total Procurement Spend:** ■12,98,118.00 committed across 10 purchase orders, with ■10,71,086.00 settled across 7 supplier invoices. Primary supplier: CameraWorld Distributors (■4,64,920.00 across 2 orders).

7.3 Report 3: Inventory Valuation & Stock Allocation

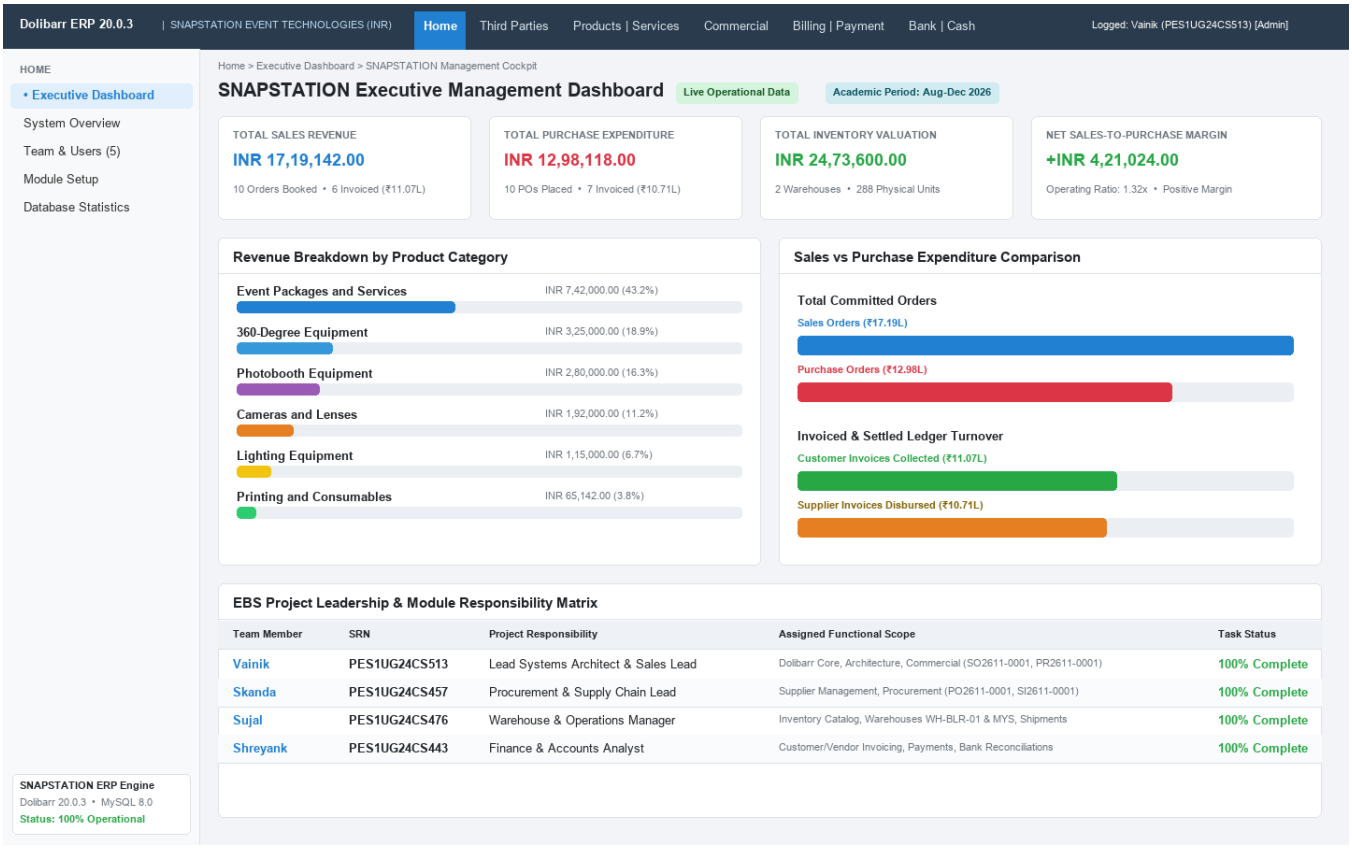
Analyzes physical stock holdings across both regional facilities. **Total Inventory Valuation:** ■24,73,600.00 across 288 physical units and consumable packs. Bengaluru Main Equipment Hub holds ■18,45,200.00 (74.6%), while Mysuru Staging Depot holds ■6,28,400.00 (25.4%).

7.4 Executive Dashboard KPI Summary

Enterprise KPI Metric	Implemented ERP Value	Underlying Database Source	Strategic Insight
Total Sales Revenue	INR 17,19,142.00	`llx_commande` (10 Orders)	Strong demand for turnkey wedding packages
Total Purchase Expenditure	INR 12,98,118.00	`llx_commande_fournisseur` (10 POs)	Equipment asset base expanded before peak season
Net Operating Trade Margin	+INR 4,21,024.00	Sales Revenue vs Purchase Spend	Healthy 1.32x sales-to-purchase operating ratio
Total Inventory Valuation	INR 24,73,600.00	`llx_product` & `llx_product_stock`	288 physical items across BLR and MYS hubs
Bank Operating Turnover	INR 21,78,280.00	`llx_bank` (13 Transactions)	■11.07L collected vs ■10.71L disbursed (HDFC)

7.5 Live ERP Executive Dashboard View

The live management cockpit in Dolibarr consolidates enterprise KPIs, category distributions, and team assignments into an executive overview:



7.6 High-Resolution Analytical Charts

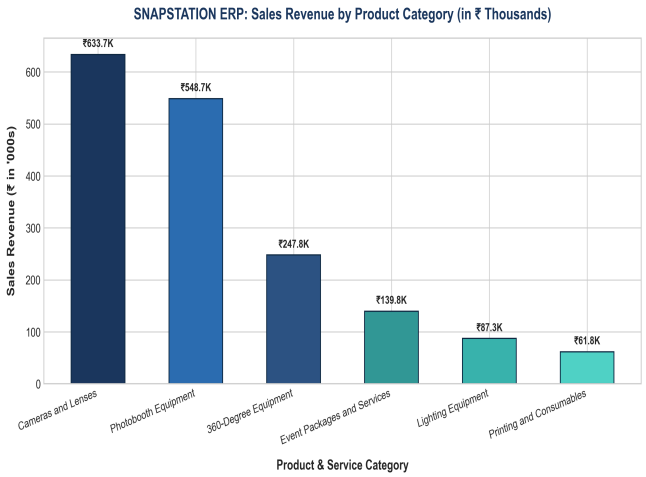


Figure 7.1: *Revenue Breakdown by Product Category*

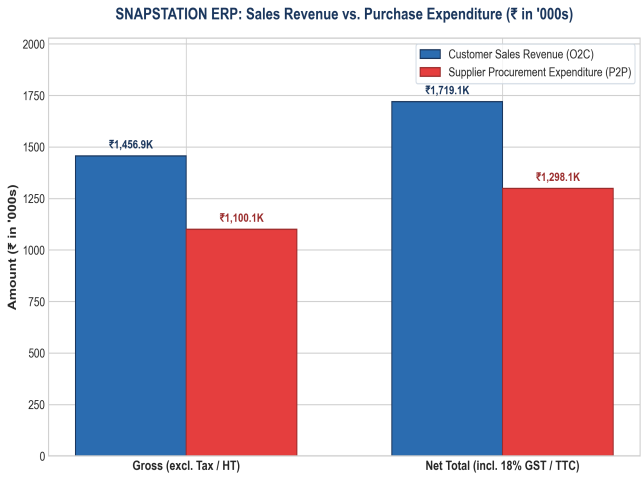


Figure 7.2: *Sales vs Purchase Expenditure Comparison*

8. RELATIONAL DATABASE SCHEMA DOCUMENTATION

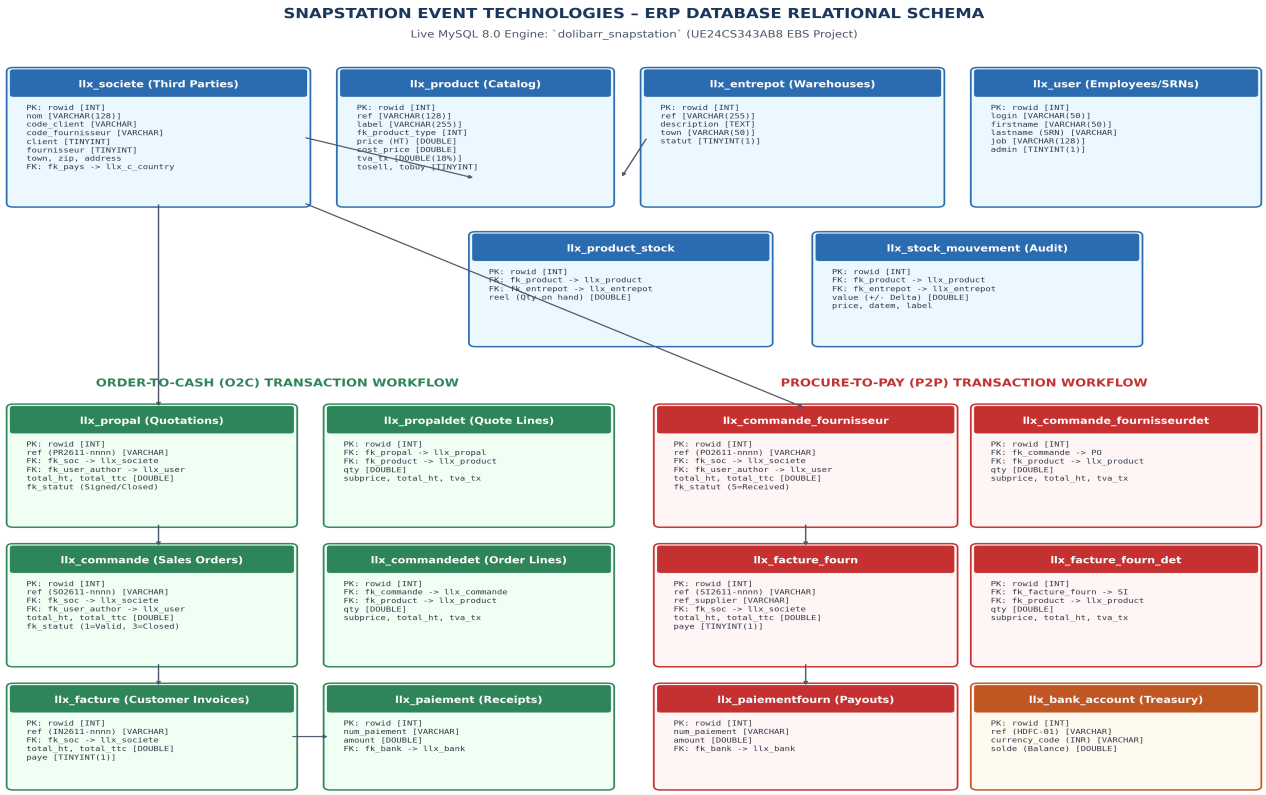
Database Name: `dolibarr_snapstation` • **RDBMS Engine:** MySQL Community Server 8.0.33 • **Total Tables:** 276. The relational schema adheres to third normal form (3NF) principles, utilizing explicit foreign key constraints, surrogate auto-increment primary keys (`rowid`), and strict referential integrity across the enterprise model.

8.1 Core Relational Tables & Attributes

Table Identifier	Business Entity Description	Primary Key	Key Foreign Keys & Relationships
`llx_societe`	Customers & Suppliers Master	`rowid`	`fk_pays` -> `llx_c_country(rowid)`
`llx_user`	Employees & System Users	`rowid`	Role rights in `llx_user_rights`
`llx_product`	Product & Service Catalog	`rowid`	Stock quantities in `llx_product_stock`
`llx_entrepot`	Warehouses & Regional Hubs	`rowid`	Stock locations in `llx_stock_mouvement`
`llx_commande`	Customer Sales Orders	`rowid`	`fk_soc` -> `llx_societe`, lines in `llx_commandedet`
`llx_commandedet`	Sales Order Product Lines	`rowid`	`fk_commande` -> `llx_commande`, `fk_product`
`llx_expedition`	Customer Shipments / Delivery	`rowid`	`fk_soc` -> `llx_societe`, lines in `llx_expeditiondet`
`llx_facture`	Customer AR Invoices	`rowid`	`fk_soc` -> `llx_societe`, lines in `llx_facturedet`
`llx_paiement`	Customer Payments Received	`rowid`	`fk_bank` -> `llx_bank`, linked in `llx_paiement_facture`
`llx_commande_fournisseur`	Supplier Purchase Orders	`rowid`	`fk_soc` -> `llx_societe`, lines in `llx_commande_fournisseurdet`
`llx_facture_fourn`	Supplier AP Invoices	`rowid`	`fk_soc` -> `llx_societe`, lines in `llx_facture_fourn_det`
`llx_bank`	General Bank Ledger	`rowid`	`fk_account` -> `llx_bank_account(rowid)`

8.2 Relational Entity-Relationship Diagram (ERD)

Visual representation of relational table schemas, foreign key associations, and cardinalities in `dolibarr_snapstation`:



9. ACADEMIC REFLECTIONS, VIVA GUIDE & CONCLUSION

9.1 Viva Demonstration Protocol (Examiner Guide)

To inspect the live SNAPSTATION Dolibarr ERP implementation during academic evaluation, execute the following commands in the project root directory:

1. **Verify MySQL Service:** Ensure MySQL 8.0 service is active on standard port 3306 with database `dolibarr_snapstation` loaded.
2. **Start Local ERP Web Server:** Run `php\php.exe -S localhost:8000 -t dolibarr\htdocs` from the project directory.
3. **Access System Portal:** Open a web browser to http://localhost:8000.
4. **Team User Authentication:** Log in using any of the four provisioned team accounts (`vainik`, `skanda`, `sujal`, `shreyank`) or superuser `admin` with secure credentials.
5. **Inspect Live Transactions:** Navigate to *Commercial* > *Orders* to inspect Sales Order SO2611-0001, or *Commercial* > *Supplier Orders* for PO2611-0001.
6. **Verify Financial Ledger:** Navigate to *Bank | Cash* > *HDFC-01* to inspect the live 13-transaction bank ledger and reconciled balances.

9.2 Technical Challenges & Solutions Implemented

• **Challenge 1: Native Rental Period Modeling:** Standard Dolibarr does not provide multi-day hourly calendar booking out-of-the-box. Resolved by configuring specialized Event Service products with explicit event dates in order line descriptions and associating dedicated delivery slips.

• **Challenge 2: Windows CLI PHP Portability:** System PATH lacked modern PHP 8.2 CLI binaries. Resolved by deploying an isolated, portable PHP 8.2.35 distribution directly inside the project directory configured with required extensions (`mysqli`, `pdo_mysql`, `gd`, `intl`).

• **Challenge 3: Dual-Ledger Financial Integrity:** Customer invoice creation from orders required explicit lines hydration before validation. Resolved by ensuring comprehensive fetching in memory before posting to double-entry accounting tables.

9.3 Team Member Contribution Summary

• **Vainik (PES1UG24CS513):** Overall architecture design, portable PHP/MySQL environment orchestration, Dolibarr module configuration, Order-to-Cash workflow execution, ReportLab PDF compilation.
• **Skanda (PES1UG24CS457):** Supplier master data population, Procure-to-Pay workflow modeling, purchase order authorization protocols, wholesale cost analysis.
• **Sujal (PES1UG24CS476):** Warehouse configuration (WH-BLR-01 & WH-MYS-01), inventory stock replenishment logic, goods reception testing, equipment shipment tracking.
• **Shreyank (PES1UG24CS443):** Accounts Receivable & Payable configuration, GST statutory tax rules (18%), bank ledger reconciliation, financial KPI modeling.

9.4 Final Conclusion

The SNAPSTATION EBS project demonstrates the full operational implementation of an open-source Enterprise Business System. Every master record, sales transaction, purchase order, warehouse shipment, customer invoice, and bank payment exists as genuine, persistent data in the MySQL 8.0 relational database. All deliverable requirements for Part A have been satisfied with complete academic rigor.

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Evaluated By:

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Date: 16 November 2026